

MACHINE LEARNING (25SC2107E)**Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No - 9	Team No - 14	Project Title - Box-Office Revenue Prediction Using Elastic Net Regression
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Abstract

The **Box-Office Revenue Prediction Using Elastic Net Regression** project is designed to predict how much money a movie may earn from its box-office performance. The success of a movie can depend on many factors, such as its budget, genre, ratings, popularity, release year, and other movie-related information. Understanding these factors can help us estimate the possible revenue of a movie.

In this project, we use a dataset containing information about different movies. The data is first collected and organized so that it can be easily understood and used. The dataset is then checked for missing or incorrect values and prepared for further analysis. Different movie features are studied to understand how they may be related to box-office revenue. Simple data analysis and visualizations can also be used to identify useful patterns in the data.

After preparing the data, a machine-learning model is used to learn from the information available in previous movies. The model takes important movie details as input and gives an estimated box-office revenue as the output. The predicted revenue can then be compared with the actual revenue to understand how well the prediction works.

The main goal of this project is to show how movie-related data can be used to make revenue predictions. It provides a simple example of applying machine learning to a real-world problem in the entertainment industry. The project can help students understand the basic steps involved in a machine-learning project, including collecting data, cleaning data, analyzing information, training a model, and making predictions.

1. Team Details

Roll Number(s)	Name of the Student	Signature
2520030525	Yeshwanth	
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2. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

The Following tools and libraries are planned to be used for developing the project:

Python – Main programming language used for data processing, analysis, and machine learning.

Jupyter Notebook – Used to write, execute, and organize Python code step by step.

Pandas – Used for loading, cleaning, organizing, and analyzing the movie dataset.

NumPy – Used for numerical calculations and handling data in array format.

Matplotlib – Used to create graphs and visualizations for understanding the data.

Seaborn – Used to create simple and informative statistical visualizations.

Scikit-learn – Used for building and evaluating the machine-learning model.

CSV Dataset – Used as the main source of movie-related data for analysis and prediction.

Guide Name and Signature (remarks if any) _____